Rockbusters ERD:

The Rockbuster database uses a snowflake schema.

The Rockbuster database follows a snowflake schema because dimension tables are normalized and split into multiple related tables, rather than being stored in a single denormalized dimension.

For example:

- The customer table links to address, which links to city, which links to country
- The film table links to language, category, and actor through bridge tables

In a star schema, all descriptive attributes would typically be stored in one wide dimension table. In contrast, Rockbuster's structure reduces redundancy by normalizing dimensions, which is characteristic of a snowflake schema.

Fact Tables – There are two fact tables in this schema:

payment (FACT TABLE)

Column	Data Type	Description
payment_id	integer	Unique identifier for each payment
customer_id	integer	Customer who made the payment
staff_id	integer	Staff member who processed the payment
rental_id	integer	Rental associated with the payment
amount	numeric	Payment amount in dollars
payment_date	timestamp	Date and time of payment

rental (FACT TABLE)

Column	Data Type	Description
rental_id	integer	Unique identifier for each rental
rental_date	timestamp	Date and time the rental began
inventory_id	integer	Inventory item rented
customer_id	integer	Customer who rented the film
return_date	timestamp	Date and time the film was returned
staff_id	integer	Staff member who handled the rental
last_update	timestamp	Last update timestamp

Dimension Tables

film

Column	Data Type	Description
film_id	integer	Unique film identifier
title	varchar	Film title
description	text	Film description
release_year	integer	Year of release
language_id	integer	Language of the film
rental_duration	integer	Allowed rental period (days)
rental_rate	numeric	Rental price
length	integer	Film duration (minutes)
replacement_cost	numeric	Cost to replace the film
rating	varchar	MPAA rating
last_update	timestamp	Last update timestamp
special_features	text	Special features included
fulltext	tsvector	Searchable text vector

actor

Column	Data Type	Description
actor_id	integer	Unique actor identifier
first_name	varchar	Actor’s first name
last_name	varchar	Actor’s last name
last_update	timestamp	Last update timestamp

film_actor (bridge table)

Column	Data Type	Description
actor_id	integer	Actor identifier
film_id	integer	Film identifier
last_update	timestamp	Last update timestamp

category

Column	Data Type	Description
category_id	integer	Category identifier
name	varchar	Film category
last_update	timestamp	Last update timestamp

film_category (bridge table)

Column	Data Type	Description
film_id	integer	Film identifier
category_id	integer	Category identifier
last_update	timestamp	Last update timestamp

customer

Column	Data Type	Description
customer_id	integer	Unique customer identifier
store_id	integer	Store associated with customer
first_name	varchar	Customer first name
last_name	varchar	Customer last name

Column	Data Type	Description
email	varchar	Customer email
address_id	integer	Address identifier
active	boolean	Active customer indicator
create_date	date	Account creation date
last_update	timestamp	Last update timestamp

inventory

Column	Data Type	Description
inventory_id	integer	Inventory identifier
film_id	integer	Film identifier
store_id	integer	Store identifier
last_update	timestamp	Last update timestamp

staff

Column	Data Type	Description
staff_id	integer	Staff identifier
first_name	varchar	Staff first name
last_name	varchar	Staff last name
address_id	integer	Address identifier
email	varchar	Staff email
store_id	integer	Store identifier
active	boolean	Active status
username	varchar	Login username

Column	Data Type	Description
password	varchar	Login password
last_update	timestamp	Last update timestamp
picture	bytea	Staff photo

address

Column	Data Type	Description
address_id	integer	Address identifier
address	varchar	Street address
district	varchar	District
city_id	integer	City identifier
postal_code	varchar	Postal code
phone	varchar	Phone number
last_update	timestamp	Last update timestamp

city

Column	Data Type	Description
city_id	integer	City identifier
city	varchar	City name
country_id	integer	Country identifier
last_update	timestamp	Last update timestamp

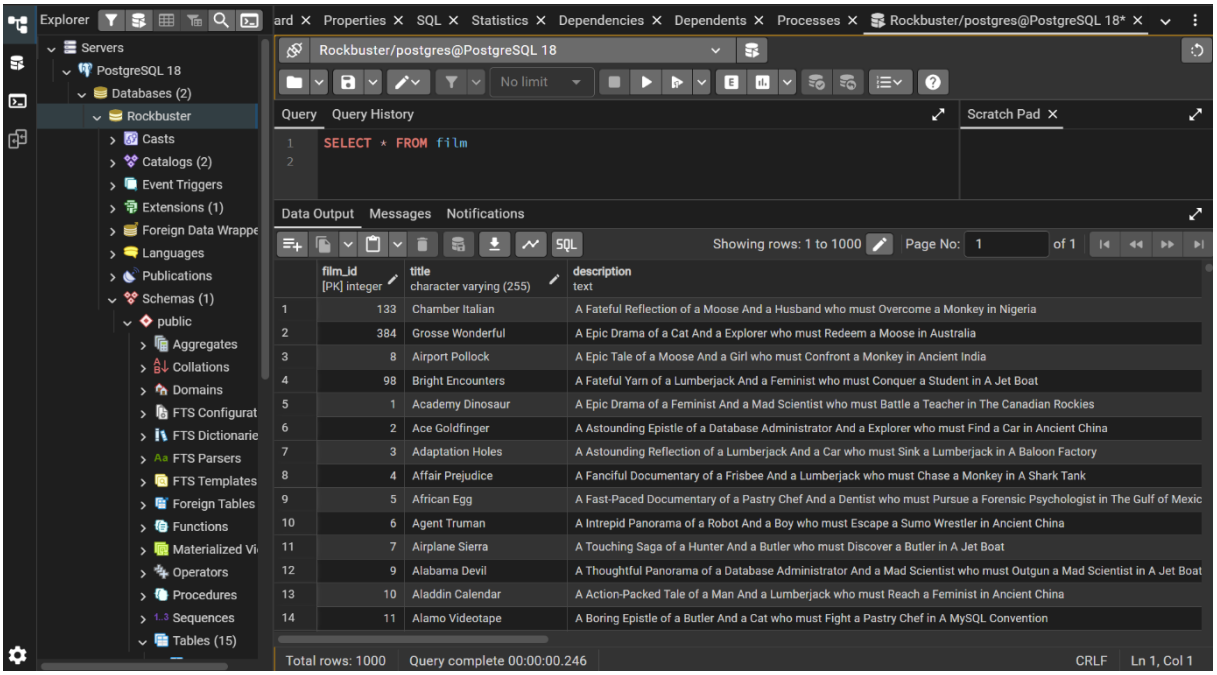
country

Column	Data Type	Description
country_id	integer	Country identifier
country	varchar	Country name
last_update	timestamp	Last update timestamp

language

Column	Data Type	Description
language_id	integer	Language identifier
name	varchar	Language name
last_update	timestamp	Last update timestamp

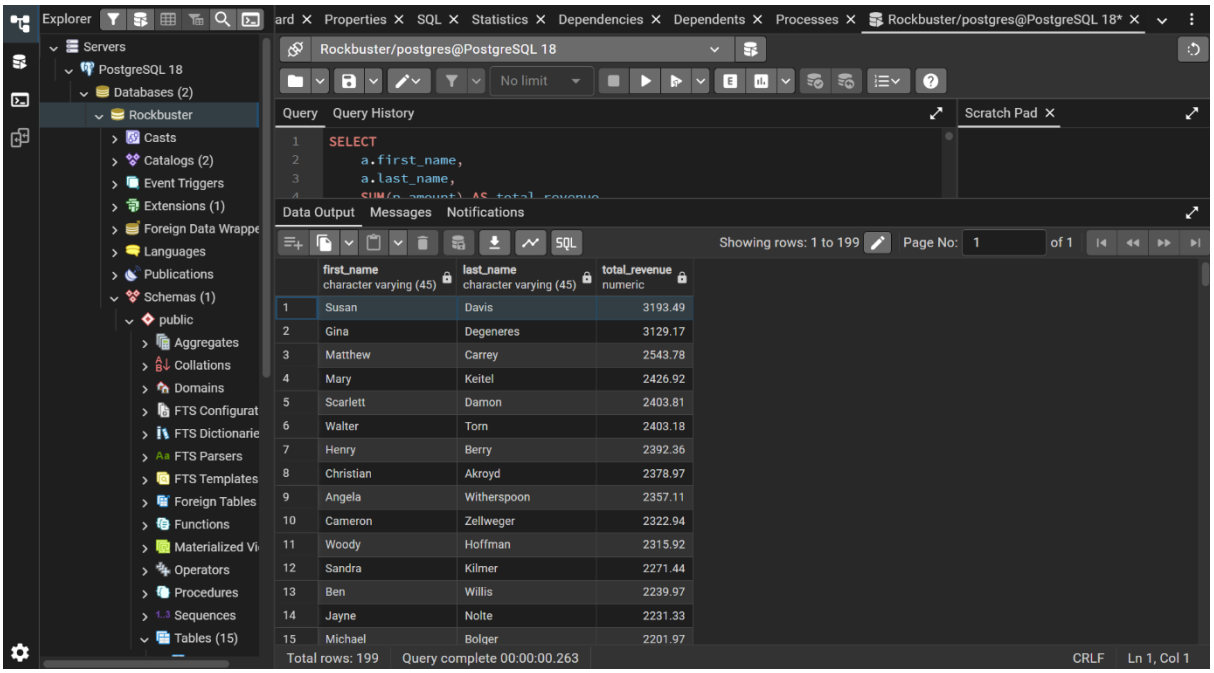
SELECT * FROM film would return the film table, for example:



Step 4: Business Questions

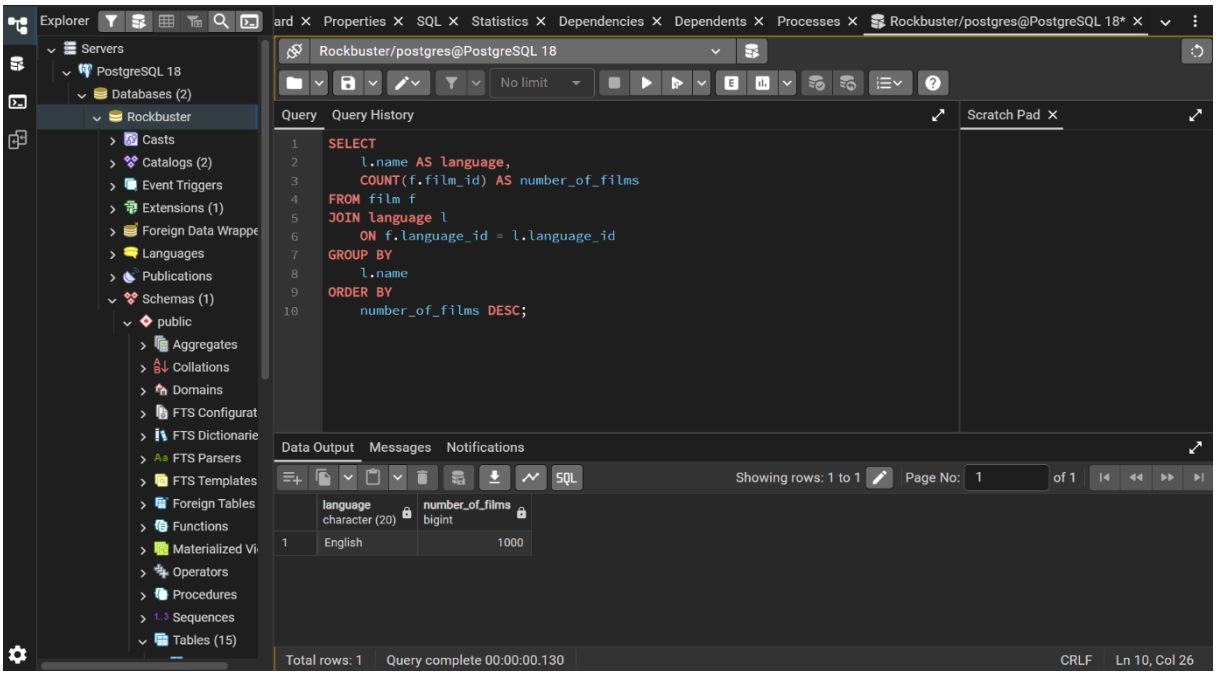
Question 1: Which actors brought Rockbuster the most revenue?

Revenue is recorded in the **payment** table, but actors are not directly linked to payments. Multiple joins are required to trace revenue from payments → rentals → films → actors. I used the help of chatgpt to get the SQL code to extract this information. The answer is Susan Davis, Gina Degeneres and Mathew Carrey brought Rochbusters the most revenue. Please look at the screenshot below.



Question 2: What language are the majority of movies in the collection?

Each film has a language identifier. Joining film to language allows counting films by language to determine which language appears most frequently. Analysis of the film and language tables shows that English is the dominant language in Rockbuster's movie collection, accounting for the largest share of available titles.


Explorer
Servers
PostgreSQL 18
Databases (2)
Rockbuster
Casts
Catalogs (2)
Event Triggers
Extensions (1)
Foreign Data Wrappe
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurat
FTS Dictionarie
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Vi
Operators
Procedures
Sequences
Tables (15)
Properties
SQL
Statistics
Dependencies
Dependents
Processes
Rockbuster/postgres@PostgreSQL 18*
Rockbuster/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
SELECT
l.name AS language,
COUNT(f.film_id) AS number_of_films
FROM film f
JOIN language l
ON f.language_id = l.language_id
GROUP BY
l.name
ORDER BY
number_of_films DESC;
Data Output
Messages
Notifications
Showing rows: 1 to 1
Page No: 1
of 1
language character (20)
number_of_films bigint
1
English
1000
Total rows: 1
Query complete 00:00:00.130
CRLF
Ln 10, Col 26